

The Full Asymptotic Transseries of the Bell Numbers

Finite all-orders saddle coefficients, Lambert– W geometry, Touchard cumulants, and polynomial–logarithmic re-expansion

A repository-guided research synthesis with complete coefficient formulae

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Abstract

The Bell numbers $\mathcal{B}_n$, defined by

$$\exp(e^z - 1) = \sum_{n \geq 0} \mathcal{B}_n \frac{z^n}{n!},$$

have a classical saddle point at $r = W_0(n)$. The leading Moser–Wyman–de Bruijn approximation is well known, but an actually usable “all-orders formula” requires more than unnamed correction polynomials: it must specify a genuine asymptotic scale, a finite rule for every coefficient, and a remainder theorem. This article supplies that data.

Write

$$\mathcal{T}_j(r) = e^{-r} (r \partial_r)^j e^r = \sum_{k=1}^j \left\{ \begin{matrix} j \\ k \end{matrix} \right\} r^k, \quad q_j(r) = \frac{\mathcal{T}_j(r)}{r(1+r)^{j/2}}.$$

For every $m \geq 0$, the m -th saddle coefficient is given by a finite partition sum

$$c_m(r) = \sum_{\substack{k_1, \dots, k_{2m} \geq 0 \\ \sum_{s=1}^{2m} s k_s = 2m}} (-1)^{m+K} (2m + 2K - 1)!! \prod_{s=1}^{2m} \frac{q_{s+2}(r)^{k_s}}{((s+2)!)^{k_s} k_s!}, \quad K = \sum_{s=1}^{2m} k_s,$$

with $c_0 = 1$. Equivalently, the same coefficients are Gaussian averages of complete exponential Bell polynomials. We prove, uniformly to arbitrary fixed order,

$$\mathcal{B}_n = \frac{n! \exp(e^r - 1)}{r^n \sqrt{2\pi n(1+r)}} \left(\sum_{m=0}^{M-1} \frac{c_m(r)}{n^m} + \mathcal{O}_M((r/n)^M) \right).$$

Every c_m is rational in r , more precisely $c_m(r) = P_m(r)/(1+r)^{3m}$ with $P_m \in \mathbb{Q}[r]$, $\deg P_m \leq 4m$, and $c_m(r) = \mathcal{O}(r^m)$.

After combining this saddle expansion with the complete Stirling series for $n!$, the Bell numbers acquire the natural exponential transseries scale $e^{-r} = r/n$:

$$\log \mathcal{B}_n \sim n \left(r - 1 + \frac{1}{r} \right) - 1 - \frac{1}{2} \log(1+r) + \sum_{m \geq 1} D_m(r) e^{-mr},$$

where every $D_m(r)$ is a bounded rational function given by finite coefficient operations. Exponentiating yields a multiplicative transseries with finite general formulae for its coefficients. Finally, substituting the Stirling-cycle-number expansion of $W_0(n)$ gives the full polynomial–logarithmic form in $L = \log n$ and $\lambda = \log L$, including the beyond-all-orders sectors $n^{-m} L^m \mathbb{Q}[\lambda][[L^{-1}]]$. The same universal saddle coefficients also govern the Touchard polynomials $\mathcal{B}_n(x)$ for fixed $x > 0$.

Scope and status

This is a self-contained mathematical article, not a claim of Lean formalization. Its principal contribution is an explicit organization of the standard saddle-point mechanism into finite coefficient formulae and a ranked transseries. The first nontrivial correction agrees with the independently reported 2026 expansion of Hwang, Li, and Zacharovas. The section on nonprincipal complex saddles is deliberately labeled *formal/hyperasymptotic*: determining all Stokes multipliers of the original Cauchy contour is a separate global steepest-descent problem.

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1 Introduction and statement of the problem

1.1 Bell numbers and the missing all-orders data

The Bell number $\mathcal{B}_n$ counts set partitions of an n -element labelled set. Its most useful analytic definition is the exponential generating function

$$F(z) := \exp(e^z - 1) = \sum_{n \geq 0} \mathcal{B}_n \frac{z^n}{n!}. \quad (1)$$

The coefficient is therefore

$$\frac{\mathcal{B}_n}{n!} = \frac{1}{2\pi i} \int_{|z|=\rho} \frac{\exp(e^z - 1)}{z^{n+1}} dz, \quad \rho > 0. \quad (2)$$

The positive saddle is selected by

$$re^r = n, \quad r = W_0(n). \quad (3)$$

This gives the classical leading approximation

$$\mathcal{B}_n \sim \frac{n! \exp(e^r - 1)}{r^n \sqrt{2\pi n(1+r)}} \sim \frac{\exp(n(r - 1 + r^{-1}) - 1)}{\sqrt{1+r}}. \quad (4)$$

The two right-hand expressions in Equation (4) are asymptotically equivalent; the second uses only the leading factor in Stirling's formula for $n!$.

A complete refinement must answer four questions.

- (i) What is the asymptotic scale?
- (ii) How is the coefficient of every term defined by a finite computation?
- (iii) What algebraic structure do those coefficients possess?
- (iv) What remainder remains after an arbitrary fixed truncation?

The linked coefficient-calculus manuscript explicitly identifies these as the missing obligations behind a schematic Moser–Wyman display. The purpose here is to discharge them for the Bell numbers and, at almost no extra cost, for the Touchard polynomials.

1.2 Three equivalent notions of “full expansion”

There are three useful normal forms, and confusing them obscures the hierarchy of small terms.

Saddle normal form. Keep $r = W_0(n)$ exact and expand the normalized Cauchy integral in powers of n^{-1} . Its coefficients grow like r^m , so the actual term of order m is $\mathcal{O}((r/n)^m)$.

Lambert–W exponential normal form. Since $r/n = e^{-r}$, absorb the Stirling series for $n!$ and rewrite the answer in powers of the genuine transmonomial e^{-r} . The coefficient functions are then bounded rational functions of r .

Polynomial–logarithmic normal form. Write $L = \log n$, $\lambda = \log L$, expand $r = W_0(n)$ in $L^{-1}\mathbb{Q}[\lambda]$, and compose. The logarithm of $\mathcal{B}_n$ splits into three ranked sectors:

$$nL \mathbb{Q}[\lambda][[L^{-1}]], \quad \log L + \mathbb{Q}[\lambda][[L^{-1}]], \quad \sum_{m \geq 1} n^{-m} L^m \mathbb{Q}[\lambda][[L^{-1}]].$$

The last family is beyond every power of L^{-1} relative to the first two. Nonprincipal complex saddles lie farther beyond all these sectors.

1.3 Road map

Section 2 develops the exact integral and the Touchard-polynomial cumulants. Section 3 proves the all-orders saddle theorem. Section 4 gives four equivalent finite formulae for the coefficients and proves their rationality. Section 5 lists the first coefficients. Section 6 combines the result with Stirling’s series and produces the e^{-r} -transseries. Section 7 performs the full polynomial–logarithmic re-expansion. Section 8 describes the formal secondary-saddle sectors, Section 9 gives the Touchard extension, Section 10 presents exact algorithms, and Section 11 quantifies the numerical improvement.

2 Exact representations and saddle geometry

2.1 Combinatorial and probabilistic formulae

The Stirling numbers of the second kind satisfy

$$\mathcal{B}_n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\}.$$

Dobinski’s formula gives a second exact representation:

$$\mathcal{B}_n = e^{-1} \sum_{k=0}^{\infty} \frac{k^n}{k!}. \quad (5)$$

Indeed,

$$e^{-1} \sum_{k \geq 0} \frac{e^{kz}}{k!} = e^{-1} \exp(e^z) = \exp(e^z - 1),$$

and comparison with Equation (1) proves Equation (5). Thus $\mathcal{B}_n$ is also the n -th raw moment of a Poisson random variable of mean one.

The saddle method may be applied either to Equation (2) or, after a local central-limit analysis, to Equation (5). The contour route is more convenient for obtaining every correction coefficient in one uniform formula.

2.2 The circular integral through the positive saddle

Set $z = re^{i\theta}$ in Equation (2), where $r = W_0(n)$. Since $dz = iz d\theta$,

$$\frac{\mathcal{B}_n}{n!} = \frac{\exp(e^r - 1)}{2\pi r^n} \int_{-\pi}^{\pi} \exp(\Phi_r(\theta)) d\theta, \quad (6)$$

where

$$\Phi_r(\theta) := e^{re^{i\theta}} - e^r - in\theta. \quad (7)$$

The saddle equation makes $\Phi'_r(0) = 0$.

The complex phase in the original z -plane is

$$\Psi_n(z) := e^z - 1 - n \operatorname{Log} z. \quad (8)$$

Its saddle equation $\Psi'_n(z) = 0$ is $ze^z = n$. Hence all formal saddles are

$$r_k = W_k(n), \quad k \in \mathbb{Z},$$

but only $r_0 = r > 0$ is needed for the ordinary Poincaré expansion on the positive real axis.

2.3 Touchard polynomials as phase cumulants

Definition 2.1 (Touchard polynomial). For $j \geq 0$, define

$$\mathcal{T}_j(r) := e^{-r} (r \partial_r)^j e^r. \quad (9)$$

Then $\mathcal{T}_0(r) = 1$, and for $j \geq 1$,

$$\mathcal{T}_j(r) = \sum_{k=1}^j \left\{ \begin{matrix} j \\ k \end{matrix} \right\} r^k. \quad (10)$$

Proof. The operator identity

$$(r \partial_r)^j = \sum_{k=0}^j \left\{ \begin{matrix} j \\ k \end{matrix} \right\} r^k \partial_r^k$$

is the standard change from powers to falling factorials. Applying it to e^r , for which $\partial_r^k e^r = e^r$, gives Equation (10). $\square$

The first values are

$$\begin{aligned} \mathcal{T}_1 &= r, \\ \mathcal{T}_2 &= r + r^2, \\ \mathcal{T}_3 &= r + 3r^2 + r^3, \\ \mathcal{T}_4 &= r + 7r^2 + 6r^3 + r^4, \\ \mathcal{T}_5 &= r + 15r^2 + 25r^3 + 10r^4 + r^5. \end{aligned}$$

They obey the differential recurrence

$$\mathcal{T}_{j+1}(r) = r(\mathcal{T}_j(r) + \mathcal{T}'_j(r)). \quad (11)$$

Lemma 2.2 (Exact phase expansion). For $|\theta|$ sufficiently small,

$$\Phi_r(\theta) = \sum_{j \geq 2} \frac{i^j e^r \mathcal{T}_j(r)}{j!} \theta^j. \quad (12)$$

In particular, with

$$A := e^r \mathcal{T}_2(r) = n(1 + r), \quad (13)$$

we have $\Phi_r(\theta) = -A\theta^2/2 + \mathcal{O}(nr^2|\theta|^3)$.

Proof. Differentiation with respect to θ acts on $e^{re^{i\theta}}$ as $ir \partial_r$ at $\theta = 0$. Therefore its j -th derivative there is $i^j e^r \mathcal{T}_j(r)$. The linear term is $ie^r \mathcal{T}_1(r)\theta = ire^r \theta = in\theta$ and cancels the last term in Equation (7). The quadratic identity follows from $\mathcal{T}_2 = r(1 + r)$ and $re^r = n$. $\square$

2.4 Standardized cumulants

Scale

$$\theta = \frac{u}{\sqrt{A}} = \frac{u}{\sqrt{n(1+r)}}. \quad (14)$$

For $s \geq 1$, define

$$q_{s+2}(r) := \frac{\mathcal{T}_{s+2}(r)}{r(1+r)^{(s+2)/2}}, \quad a_s(u; r) := \frac{i^{s+2} q_{s+2}(r)}{(s+2)!} u^{s+2}. \quad (15)$$

Then Equation (12) becomes the exact formal identity

$$\Phi_r\left(\frac{u}{\sqrt{A}}\right) = -\frac{u^2}{2} + \sum_{s \geq 1} n^{-s/2} a_s(u; r). \quad (16)$$

Since $\deg \mathcal{T}_{s+2} = s+2$,

$$q_{s+2}(r) = \mathcal{O}_s(r^{s/2}) \quad (r \rightarrow \infty). \quad (17)$$

Thus the effective small parameter is not merely $n^{-1/2}$, but

$$\varepsilon := \sqrt{\frac{r}{n}} = e^{-r/2}.$$

The Gaussian symmetry removes odd powers of ε ; the final expansion therefore proceeds in powers of $\varepsilon^2 = r/n = e^{-r}$.

3 The all-orders saddle theorem

3.1 Localization of the Cauchy integral

The first issue is not formal expansion but localization: the part of the circle away from $\theta = 0$ must be smaller than every retained algebraic term.

Lemma 3.1 (Central and complementary arcs). *There exist absolute constants $\delta, c_1, c_2 > 0$ such that, for all sufficiently large n , with $r = W_0(n)$,*

$$\operatorname{Re} \Phi_r(\theta) \leq -c_1 A \theta^2, \quad |\theta| \leq \delta/r, \quad (18)$$

$$\operatorname{Re} \Phi_r(\theta) \leq -c_2 n/r^2, \quad \delta/r \leq |\theta| \leq \pi. \quad (19)$$

Consequently the complementary arc contributes

$$\int_{\delta/r \leq |\theta| \leq \pi} \exp(\Phi_r(\theta)) d\theta = \mathcal{O}(\exp(-c_2 n/r^2)), \quad (20)$$

which is smaller than $(r/n)^M$ for every fixed M .

Proof. For the central bound, Theorem 2.2 gives

$$\operatorname{Re} \Phi_r(\theta) = -\frac{1}{2} A \theta^2 + \mathcal{O}(nr^2 |\theta|^3).$$

Because $A = n(1+r) \asymp nr$, the ratio of the error to $A\theta^2$ is $\mathcal{O}(r|\theta|)$. Choosing δ sufficiently small proves Equation (18).

For the complementary bound, discard the oscillating cosine:

$$\begin{aligned} \operatorname{Re} \Phi_r(\theta) &= e^{r \cos \theta} \cos(r \sin \theta) - e^r \\ &\leq e^{r \cos \theta} - e^r = -e^r (1 - e^{-r(1 - \cos \theta)}). \end{aligned}$$

On $[-\pi, \pi]$, $1 - \cos \theta \geq C \min(\theta^2, 1)$. If $|\theta| \geq \delta/r$, then $r(1 - \cos \theta) \geq C'\delta^2/r$ until this quantity is bounded away from zero; in either case

$$1 - e^{-r(1 - \cos \theta)} \geq C''/r.$$

Since $e^r = n/r$, this gives Equation (19). Finally, $n/r^2 \rightarrow \infty$ faster than $\log(n/r)$, so the exponential in Equation (20) is smaller than every fixed power of r/n . $\square$

Remark 3.2. The much-studied Hayman admissibility of $F(z) = \exp(e^z - 1)$ gives a general framework for this localization. The elementary inequalities above are included so that the particular Bell-number argument does not depend on quoting the full admissibility machinery.

3.2 Gaussian reduction

Let Z be standard normal and write

$$\mathbb{E}f(Z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(u) e^{-u^2/2} du.$$

After Equation (14), the central part of Equation (6) becomes

$$\frac{1}{\sqrt{2\pi A}} \int e^{-u^2/2} \exp\left(\sum_{s \geq 1} n^{-s/2} a_s(u; r)\right) du, \quad (21)$$

up to a contribution smaller than every power of r/n . Formally replacing the finite scaled interval by $\mathbb{R}$ is justified by Equation (18); the Gaussian tails are super-algebraically small.

For a precise truncation, choose $0 < \alpha < 1/6$ and restrict first to $|u| \leq U_n := (n/r)^\alpha$. On this interval,

$$n^{-s/2} a_s(u; r) = \mathcal{O}_s\left((r/n)^{s/2} U_n^{s+2}\right),$$

and the right side tends to zero for every fixed s . Taylor's theorem for the exponential may therefore be applied uniformly to any prescribed order. Outside $[-U_n, U_n]$, Equation (18) supplies a bound $\mathcal{O}(e^{-cU_n^2})$. This is the standard Watson-lemma mechanism, here with coefficients depending slowly on r .

3.3 All-orders theorem

Theorem 3.3 (Full principal-saddle expansion). *Let $r = W_0(n)$, and define a_s by Equation (15). Put*

$$c_m(r) := \mathbb{E} \left[t^{2m} \exp\left(\sum_{s \geq 1} a_s(Z; r) t^s\right) \right], \quad m \geq 0. \quad (22)$$

Then $c_0(r) = 1$, and for every fixed integer $M \geq 1$,

$$\boxed{\mathcal{B}_n = \frac{n! \exp(e^r - 1)}{r^n \sqrt{2\pi n(1+r)}} \left(\sum_{m=0}^{M-1} \frac{c_m(r)}{n^m} + \mathcal{O}_M((r/n)^M) \right)}. \quad (23)$$

The constant in the remainder depends on M , but not on n .

Proof. Expand the second exponential in Equation (21) in the bookkeeping variable $t = n^{-1/2}$:

$$\exp\left(\sum_{s \geq 1} a_s(u; r) t^s\right) = \sum_{j \geq 0} P_j(u; r) t^j. \quad (24)$$

Every monomial contributing to P_j has the form $\prod_s a_s^{k_s}$ with $\sum_s s k_s = j$. Its degree in u is

$$\sum_s (s+2)k_s = j + 2 \sum_s k_s,$$

which has the same parity as j . Therefore $P_j(-u; r) = (-1)^j P_j(u; r)$, and $\mathbb{E}P_j(Z; r) = 0$ for odd j . The coefficient of n^{-m} is exactly $c_m(r) = \mathbb{E}P_{2m}(Z; r)$.

By Equation (17), each monomial in P_{2m} is $\mathcal{O}_m(r^m)$ after Gaussian averaging, because $\sum_s s k_s = 2m$. Hence

$$c_m(r) = \mathcal{O}_m(r^m). \quad (25)$$

The uniform Taylor argument preceding the theorem, with enough terms retained, shows that the first omitted even contribution is $\mathcal{O}_M(r^M n^{-M})$, while the central Gaussian tail and the complementary arc are smaller than this by Theorem 3.1. Multiplying by the prefactor in Equation (6) yields Equation (23). $\square$

Main result

The asymptotic scale in Equation (23) is genuine:

$$1, \quad \frac{r}{n}, \quad \left(\frac{r}{n}\right)^2, \quad \dots \quad \left(\frac{r}{n} = e^{-r}\right).$$

The displayed coefficient c_m/n^m has size $\mathcal{O}((r/n)^m)$, and the remainder after M terms has the size of the next member of this scale. No unnamed correction polynomial is required.

4 Finite coefficient calculus

4.1 Complete Bell-polynomial formula

Let Y_j denote the complete exponential Bell polynomial, characterized by

$$\exp\left(\sum_{s \geq 1} x_s \frac{t^s}{s!}\right) = \sum_{j \geq 0} Y_j(x_1, \dots, x_j) \frac{t^j}{j!}. \quad (26)$$

Taking $x_s = s!a_s(Z; r)$ in Equation (26) gives the first finite closed rule.

Proposition 4.1 (Bell-polynomial coefficient). *For $m \geq 0$,*

$$c_m(r) = \frac{1}{(2m)!} \mathbb{E}[Y_{2m}(1!a_1(Z; r), 2!a_2(Z; r), \dots, (2m)!a_{2m}(Z; r))]. \quad (27)$$

Thus c_m uses only $\mathcal{T}_3, \mathcal{T}_4, \dots, \mathcal{T}_{2m+2}$.

Proof. This is coefficient comparison in Equation (26); terms with $s > 2m$ cannot affect $[t^{2m}]$. $\square$

The Gaussian average can itself be written as a differential operator:

$$\mathbb{E}P(Z) = \exp\left(\frac{1}{2}\partial_u^2\right) P(u) \Big|_{u=0} \quad (28)$$

for every polynomial P . Consequently Equation (27) is a finite composition of two standard coefficient-calculus operations: a complete Bell polynomial followed by the heat operator at the origin.

4.2 Explicit partition sum

The Bell-polynomial formula becomes completely elementary after expanding the exponential and using $\mathbb{E}Z^{2j} = (2j-1)!!$, $\mathbb{E}Z^{2j+1} = 0$.

Theorem 4.2 (Finite partition formula). *For $m \geq 1$,*

$$c_m(r) = \sum_{\substack{k_1, \dots, k_{2m} \geq 0 \\ \sum_{s=1}^{2m} sk_s = 2m}} (-1)^{m+K} (2m+2K-1)!! \prod_{s=1}^{2m} \frac{q_{s+2}(r)^{k_s}}{((s+2)!)^{k_s} k_s!}, \quad K := \sum_{s=1}^{2m} k_s. \quad (29)$$

We use the convention $(-1)!! = 1$.

Proof. The coefficient of t^{2m} in $\exp(\sum_s a_s t^s)$ is

$$\sum_{\sum sk_s = 2m} \prod_s \frac{a_s^{k_s}}{k_s!}.$$

The power of Z in such a term is $\sum (s+2)k_s = 2m+2K$, and the power of i is the same. Hence

$$i^{2m+2K} = (-1)^{m+K}, \quad \mathbb{E}Z^{2m+2K} = (2m+2K-1)!!.$$

Substitution of Equation (15) proves Equation (29). $\square$

Remark 4.3 (Finiteness). The sum is indexed by partitions of $2m$: the vector $(k_1, \dots, k_{2m})$ records k_s parts of size s . Therefore the number of summands is the partition number $p(2m)$, not an infinite or asymptotic count. This is the requested finite coefficient-extraction rule.

4.3 A dynamic recurrence

Direct enumeration of partitions is conceptually transparent. For symbolic computation, a recurrence for the intermediate polynomials is often faster. Differentiating Equation (24) with respect to t gives

$$P_0(u; r) = 1, \quad P_j(u; r) = \frac{1}{j} \sum_{s=1}^j s a_s(u; r) P_{j-s}(u; r) \quad (j \geq 1). \quad (30)$$

Then

$$c_m(r) = \mathbb{E}P_{2m}(Z; r). \quad (31)$$

Because parity is preserved by Equation (30), odd P_j need not be Gaussian-averaged at all.

4.4 Rationality, denominators, and degrees

Theorem 4.4 (Algebraic structure of the coefficients). *For every $m \geq 0$, there is a polynomial $P_m(r) \in \mathbb{Q}[r]$ such that*

$$c_m(r) = \frac{P_m(r)}{(1+r)^{3m}}, \quad \deg P_m \leq 4m. \quad (32)$$

Moreover

$$c_m(r) = \mathcal{O}_m(r^m) \quad (r \rightarrow \infty). \quad (33)$$

Proof. Fix a vector in Equation (29). Since $\sum sk_s = 2m$,

$$\prod_s (1+r)^{(s+2)k_s/2} = (1+r)^{m+K}. \quad (34)$$

Also $K \leq 2m$, because every part has size at least one. Bringing the term to the common denominator $(1+r)^{3m}$ multiplies its numerator by $(1+r)^{2m-K}$.

Now $\mathcal{T}_{s+2}(r)/r$ is a polynomial of degree $s+1$. The product of these polynomials has degree

$$\sum_s (s+1)k_s = 2m + K.$$

After multiplication by $(1+r)^{2m-K}$, the degree is at most $4m$. This proves Equation (32). Finally, $q_{s+2}(r) = \mathcal{O}(r^{s/2})$, so every summand in Equation (29) is $\mathcal{O}(r^{\sum sk_s/2}) = \mathcal{O}(r^m)$, proving Equation (33). $\square$

Corollary 4.5 (Asymptotic sequence). *For fixed m ,*

$$\frac{c_{m+1}(r)n^{-m-1}}{c_m(r)n^{-m}} = \mathcal{O}_m(r/n) = \mathcal{O}_m(e^{-r})$$

whenever the denominator is nonzero. Thus the sequence of term sizes is strictly asymptotic.

4.5 The limiting Stirling identity

The highest-degree part of c_m has a classical interpretation. Define

$$\gamma_m := \lim_{r \rightarrow \infty} \frac{c_m(r)}{r^m}. \quad (35)$$

Because $q_{s+2}(r)/r^{s/2} \rightarrow 1$, Equation (22) yields

$$\gamma_m = \mathbb{E} \left[[t^{2m}] \exp \left(\sum_{s \geq 1} \frac{i^{s+2} Z^{s+2}}{(s+2)!} t^s \right) \right]. \quad (36)$$

Comparison with the saddle derivation of Stirling's formula gives the formal identity

$$\sum_{m \geq 0} \gamma_m x^m = \exp \left(- \sum_{j \geq 1} \frac{\beta_{2j}}{2j(2j-1)} x^{2j-1} \right), \quad (37)$$

where

$$\frac{t}{e^t - 1} = \sum_{j \geq 0} \beta_j \frac{t^j}{j!}, \quad \beta_1 = -\frac{1}{2}. \quad (38)$$

This is the Gaussian coefficient identity that appears when Stirling's formula is derived on two equivalent saddle contours.

The first values are

$$\gamma_0 = 1, \quad \gamma_1 = -\frac{1}{12}, \quad \gamma_2 = \frac{1}{288}, \quad \gamma_3 = \frac{139}{51840}, \quad \gamma_4 = -\frac{571}{2488320}. \quad (39)$$

Equation [Equation \(37\)](#) supplies a useful independent check on any symbolic implementation of [Equation \(29\)](#).

5 Explicit principal-saddle coefficients

5.1 The first correction by hand

Only a_1 and a_2 can contribute to c_1 :

$$c_1 = \mathbb{E} \left(a_2(Z; r) + \frac{1}{2} a_1(Z; r)^2 \right).$$

Since

$$a_1 = -\frac{i q_3}{6} u^3, \quad a_2 = \frac{q_4}{24} u^4,$$

and $\mathbb{E} Z^4 = 3, \mathbb{E} Z^6 = 15$,

$$c_1(r) = \frac{1}{8} q_4(r) - \frac{5}{24} q_3(r)^2. \quad (40)$$

Substitution of $\mathcal{T}_3$ and $\mathcal{T}_4$ gives

$$c_1(r) = -\frac{2r^4 + 9r^3 + 16r^2 + 6r + 2}{24(1+r)^3}. \quad (41)$$

The sign is worth noting: for large r , $c_1(r) \sim -r/12$.

5.2 The second correction

At order t^4 , the partitions of four give

$$c_2 = -\frac{1}{48} q_6 + \frac{7}{48} q_3 q_5 + \frac{35}{384} q_4^2 - \frac{35}{64} q_3^2 q_4 + \frac{385}{1152} q_3^4. \quad (42)$$

After rational simplification,

$$c_2(r) = \frac{4r^8 - 12r^7 - 143r^6 - 384r^5 - 588r^4 - 636r^3 + 100r^2 + 24r + 4}{1152(1+r)^6}. \quad (43)$$

5.3 The third correction

The finite rule [Equation \(29\)](#) gives

$$c_3(r) = \frac{1}{414720(1+r)^9} \left(1112r^{12} + 12708r^{11} + 56082r^{10} + 118683r^9 \right. \\ \left. + 97176r^8 - 74394r^7 - 259046r^6 - 456444r^5 \right. \\ \left. - 635424r^4 + 115308r^3 + 39432r^2 + 10008r + 1112 \right). \quad (44)$$

The leading coefficients of c_1, c_2, c_3 agree with $\gamma_1, \gamma_2, \gamma_3$ in [Equation \(39\)](#).

5.4 How to read the first terms

Substituting Equations (41), (43) and (44) into Equation (23) gives

$$\mathcal{B}_n = \frac{n! \exp(e^r - 1)}{r^n \sqrt{2\pi n(1+r)}} \left(1 + \frac{c_1(r)}{n} + \frac{c_2(r)}{n^2} + \frac{c_3(r)}{n^3} + \mathcal{O}\left(\frac{r^4}{n^4}\right) \right). \quad (45)$$

This is already a complete Poincare expansion, but it hides the most natural transmonomial because $c_m(r)$ grows like r^m . The next section absorbs Stirling's series and rewrites every term as a bounded coefficient times e^{-mr} .

6 Logarithmic and exponential transseries

6.1 Combining the saddle series with Stirling's series

Stirling's complete logarithmic expansion is

$$\log n! \sim n \log n - n + \frac{1}{2} \log(2\pi n) + \sum_{j \geq 1} \frac{\beta_{2j}}{2j(2j-1)n^{2j-1}}. \quad (46)$$

Define

$$\sigma_m := \begin{cases} \frac{\beta_{m+1}}{m(m+1)}, & m \text{ odd}, \\ 0, & m \text{ even}. \end{cases} \quad (47)$$

Thus the last sum in Equation (46) is $\sum_{m \geq 1} \sigma_m n^{-m}$.

Let

$$C(x; r) := 1 + \sum_{m \geq 1} c_m(r) x^m, \quad \log C(x; r) = \sum_{m \geq 1} \ell_m(r) x^m. \quad (48)$$

Every ℓ_m is finite and can be obtained in either of the forms

$$\ell_m(r) = \sum_{\substack{k_1, \dots, k_m \geq 0 \\ \sum_{j=1}^m j k_j = m}} (-1)^{K+1} (K-1)! \prod_{j=1}^m \frac{c_j(r)^{k_j}}{k_j!}, \quad K = \sum_j k_j, \quad (49)$$

$$\ell_1 = c_1, \quad \ell_m = c_m - \frac{1}{m} \sum_{j=1}^{m-1} j \ell_j c_{m-j} \quad (m \geq 2). \quad (50)$$

The recurrence follows from $C' = C(\log C)'$.

Definition 6.1 (Logarithmic saddle coefficients). Set

$$d_m(r) := \sigma_m + \ell_m(r). \quad (51)$$

Theorem 6.2 (Lambert-W logarithmic form). For every fixed $M \geq 1$, with $r = W_0(n)$,

$$\log \mathcal{B}_n = n \left(r - 1 + \frac{1}{r} \right) - 1 - \frac{1}{2} \log(1+r) + \sum_{m=1}^{M-1} \frac{d_m(r)}{n^m} + \mathcal{O}_M((r/n)^M). \quad (52)$$

Equivalently,

$$\mathcal{B}_n = \frac{\exp(n(r - 1 + r^{-1}) - 1)}{\sqrt{1+r}} \exp\left(\sum_{m=1}^{M-1} \frac{d_m(r)}{n^m} + \mathcal{O}_M((r/n)^M)\right). \quad (53)$$

Each d_m has the form

$$d_m(r) = \frac{Q_m(r)}{(1+r)^{3m}}, \quad Q_m \in \mathbb{Q}[r], \quad \deg Q_m \leq 4m, \quad (54)$$

and $d_m(r) = \mathcal{O}_m(r^m)$.

Proof. Insert Equation (46) and Equation (48) into the logarithm of Equation (23). The identities

$$\log n - \log r = r, \quad e^r = \frac{n}{r}$$

follow from $re^r = n$ and produce the first line of Equation (52). The structure statement follows from Theorem 4.4, the finite logarithm formula Equation (49), and a common denominator $(1+r)^{3m}$. $\square$

6.2 The genuine e^{-r} -transseries

Define the bounded rational functions

$$D_m(r) := \frac{d_m(r)}{r^m}. \quad (55)$$

Since $n = re^r$,

$$\frac{d_m(r)}{n^m} = D_m(r)e^{-mr}.$$

Therefore Equation (52) becomes the genuine exponential transseries.

Transseries form

For every fixed M ,

$$\begin{aligned} \log \mathcal{B}_n = & n \left(r - 1 + \frac{1}{r} \right) - 1 - \frac{1}{2} \log(1+r) \\ & + \sum_{m=1}^{M-1} D_m(r)e^{-mr} + \mathcal{O}_M(e^{-Mr}), \quad r = W_0(n). \end{aligned} \quad (56)$$

Every D_m is rational and bounded as $r \rightarrow \infty$. Thus $1, e^{-r}, e^{-2r}, \dots$ is an honest asymptotic scale with bounded coefficient functions.

The first logarithmic coefficients simplify markedly:

$$d_1(r) = -\frac{r^2(2r^2 + 7r + 10)}{24(1+r)^3}, \quad (57)$$

$$d_2(r) = -\frac{r^3(2r^4 + 12r^3 + 29r^2 + 40r + 36)}{48(1+r)^6}, \quad (58)$$

$$\begin{aligned} d_3(r) = & \frac{r^4}{5760(1+r)^9} (16r^8 + 164r^7 + 616r^6 + 818r^5 - 1096r^4 \\ & - 5793r^3 - 10032r^2 - 12208r - 12048). \end{aligned} \quad (59)$$

Consequently

$$D_1(r) = -\frac{r(2r^2 + 7r + 10)}{24(1+r)^3}, \quad (60)$$

$$D_2(r) = -\frac{r(2r^4 + 12r^3 + 29r^2 + 40r + 36)}{48(1+r)^6}, \quad (61)$$

$$D_3(r) = \frac{r}{5760(1+r)^9} (16r^8 + 164r^7 + 616r^6 + 818r^5 - 1096r^4 - 5793r^3 - 10032r^2 - 12208r - 12048). \quad (62)$$

Their large- r expansions begin

$$D_1(r) = -\frac{1}{12} - \frac{1}{24r} - \frac{1}{24r^2} + \frac{1}{3r^3} - \frac{5}{6r^4} + \cdots, \quad (63)$$

$$D_2(r) = -\frac{1}{24r} + \frac{1}{48r^3} - \frac{1}{8r^4} + \cdots, \quad (64)$$

$$D_3(r) = \frac{1}{360} + \frac{1}{288r} - \frac{7}{288r^2} + \frac{7}{2880r^3} + \cdots. \quad (65)$$

Because the only finite singularities of these displayed rational functions are at $r = 0$ and $r = -1$, their inverse- r expansions converge for sufficiently large $|r|$, not merely formally.

6.3 The first correction and comparison with the literature

Keeping only d_1/n in Equation (53) gives

$$\mathcal{B}_n = \frac{\exp(n(r-1+r^{-1})-1)}{\sqrt{1+r}} \left(1 - \frac{r^2(2r^2+7r+10)}{24(1+r)^3n} + \mathcal{O}\left(\frac{r^2}{n^2}\right) \right). \quad (66)$$

This is exactly the first correction reported in the 2026 expansion of Hwang, Li, and Zacharovas, with their notation $\omega_n = W(n)$. Here it is not an isolated correction: it is the first member of the finite all-orders mechanism Equations (29), (49) and (51).

6.4 Multiplicative exponential coefficients

Define $A_0(r) = 1$ and

$$\exp\left(\sum_{j \geq 1} D_j(r) y^j\right) = \sum_{m \geq 0} A_m(r) y^m. \quad (67)$$

Then

$$A_m(r) = \sum_{\substack{k_1, \dots, k_m \geq 0 \\ \sum_{j=1}^m j k_j = m}} \prod_{j=1}^m \frac{D_j(r)^{k_j}}{k_j!}, \quad (68)$$

$$mA_m(r) = \sum_{j=1}^m j D_j(r) A_{m-j}(r). \quad (69)$$

Exponentiating Equation (56) gives

$$\mathcal{B}_n = \frac{\exp(n(r-1+r^{-1})-1)}{\sqrt{1+r}} \left(\sum_{m=0}^{M-1} A_m(r) e^{-mr} + \mathcal{O}_M(e^{-Mr}) \right). \quad (70)$$

The first three bounded coefficient functions are

$$A_1(r) = -\frac{r(2r^2+7r+10)}{24(1+r)^3}, \quad (71)$$

$$A_2(r) = \frac{r(4r^5 - 20r^4 - 199r^3 - 556r^2 - 860r - 864)}{1152(1+r)^6}, \quad (72)$$

$$A_3(r) = \frac{r}{414720(1+r)^9} (1112r^8 + 12828r^7 + 55962r^6 + 111301r^5 + 55818r^4 - 196476r^3 - 492584r^2 - 749376r - 867456). \quad (73)$$

6.5 Bernoulli boundary values

Taking the logarithm of Equation (37) identifies the limits of the bounded logarithmic coefficients.

Proposition 6.3 (Limits at the transseries boundary). *For $m \geq 1$,*

$$\lim_{r \rightarrow \infty} D_m(r) = \begin{cases} -\frac{\beta_{m+1}}{m(m+1)}, & m \text{ odd}, \\ 0, & m \text{ even}. \end{cases} \quad (74)$$

Moreover

$$\lim_{r \rightarrow \infty} A_m(r) = \gamma_m. \quad (75)$$

Proof. By Equation (48), the coefficient of x^m in $\log \sum_j c_j(r)x^j$, after the scaling $x \mapsto x/r$, tends to the coefficient of x^m in $\log \sum_j \gamma_j x^j$. Equation (37) gives Equation (74); the fixed Stirling term σ_m/r^m tends to zero. Exponentiating proves Equation (75). $\square$

7 Polynomial-logarithmic re-expansion

7.1 The all-orders Lambert-W coefficients

Set

$$L := \log n, \quad \lambda := \log L. \quad (76)$$

The principal Lambert function has the Stirling-cycle-number expansion

$$r = W_0(n) \sim L - \lambda + \sum_{N \geq 1} \frac{R_N(\lambda)}{L^N}, \quad (77)$$

where

$$R_N(\lambda) = \sum_{m=1}^N \frac{(-1)^{N-m}}{m!} \left[\begin{matrix} N \\ N-m+1 \end{matrix} \right] \lambda^m. \quad (78)$$

Here $\left[\begin{matrix} N \\ j \end{matrix} \right]$ is the unsigned Stirling cycle number. The first polynomials are

$$\begin{aligned} R_1 &= \lambda, \\ R_2 &= \frac{\lambda(\lambda - 2)}{2}, \\ R_3 &= \frac{\lambda(2\lambda^2 - 9\lambda + 6)}{6}, \\ R_4 &= \frac{\lambda(3\lambda^3 - 22\lambda^2 + 36\lambda - 12)}{12}, \\ R_5 &= \frac{\lambda(12\lambda^4 - 125\lambda^3 + 350\lambda^2 - 300\lambda + 60)}{60}. \end{aligned} \quad (79)$$

Formula Equation (78) is the exact coefficient bridge between the combinatorial first-kind Stirling calculus and the logarithmic transseries calculus.

7.2 The extensive sector: coefficients multiplying n

Put $z = L^{-1}$ and define

$$\Delta(z; \lambda) := 1 - \lambda z + \sum_{j \geq 1} R_j(\lambda) z^{j+1}. \quad (80)$$

Then $r = z^{-1} \Delta(z; \lambda)$. Define

$$\boxed{H_N(\lambda) := R_N(\lambda) + [z^{N-1}] \Delta(z; \lambda)^{-1}.} \quad (81)$$

Only $R_1, \dots, R_N$ are needed to calculate H_N . Since $1/r = z \Delta^{-1}$,

$$r - 1 + \frac{1}{r} \sim L - \lambda - 1 + \sum_{N \geq 1} \frac{H_N(\lambda)}{L^N}. \quad (82)$$

The first coefficients are

$$\begin{aligned} H_1 &= \lambda + 1, \\ H_2 &= \frac{\lambda^2}{2}, \\ H_3 &= \frac{\lambda^2(2\lambda - 3)}{6}, \\ H_4 &= \frac{\lambda^2(3\lambda^2 - 10\lambda + 6)}{12}, \\ H_5 &= \frac{\lambda^2(12\lambda^3 - 65\lambda^2 + 90\lambda - 30)}{60}, \\ H_6 &= \frac{\lambda^2(20\lambda^4 - 154\lambda^3 + 355\lambda^2 - 280\lambda + 60)}{120}. \end{aligned} \quad (83)$$

Therefore the classical de Bruijn expansion is refined to arbitrary order by

$$\frac{\log \mathcal{B}_n}{n} \sim L - \lambda - 1 + \sum_{N \geq 1} \frac{H_N(\lambda)}{L^N}, \quad (84)$$

where all omitted sectors are $\mathcal{O}(\log L/n)$, hence beyond every power of L^{-1} .

Written out through L^{-6} ,

$$\begin{aligned} \frac{\log \mathcal{B}_n}{n} &\sim L - \lambda - 1 + \frac{\lambda + 1}{L} + \frac{\lambda^2}{2L^2} + \frac{\lambda^2(2\lambda - 3)}{6L^3} \\ &\quad + \frac{\lambda^2(3\lambda^2 - 10\lambda + 6)}{12L^4} + \frac{\lambda^2(12\lambda^3 - 65\lambda^2 + 90\lambda - 30)}{60L^5} \\ &\quad + \frac{\lambda^2(20\lambda^4 - 154\lambda^3 + 355\lambda^2 - 280\lambda + 60)}{120L^6} + \dots \end{aligned} \quad (85)$$

7.3 The edge sector

The factor $(1 + r)^{-1/2}$ lives on a lower transseries level than the terms multiplied by n . Define

$$\Sigma(z; \lambda) := z(1 + r) = \Delta(z; \lambda) + z = 1 + (1 - \lambda)z + \sum_{j \geq 1} R_j(\lambda) z^{j+1}, \quad (86)$$

and

$$\boxed{E_N(\lambda) := -\frac{1}{2}[z^N] \log \Sigma(z; \lambda).} \quad (87)$$

Then

$$-1 - \frac{1}{2} \log(1+r) \sim -1 - \frac{1}{2} \log L + \sum_{N \geq 1} \frac{E_N(\lambda)}{L^N}. \quad (88)$$

The first values are

$$\begin{aligned} E_1 &= \frac{\lambda - 1}{2}, \\ E_2 &= \frac{\lambda^2 - 4\lambda + 1}{4}, \\ E_3 &= \frac{2\lambda^3 - 15\lambda^2 + 18\lambda - 2}{12}, \\ E_4 &= \frac{3\lambda^4 - 34\lambda^3 + 84\lambda^2 - 48\lambda + 3}{24}, \\ E_5 &= \frac{12\lambda^5 - 185\lambda^4 + 730\lambda^3 - 900\lambda^2 + 300\lambda - 12}{120}. \end{aligned} \quad (89)$$

As with ℓ_m , an explicit partition formula follows by expanding $\log(1+u)$; Equation (87) is usually the shortest exact definition.

7.4 All exponentially small sectors

For each $m \geq 1$, define polynomial-logarithmic coefficient polynomials $p_{m,j}(\lambda)$ by

$$p_{m,j}(\lambda) := [z^j] \left\{ z^m d_m(z^{-1} \Delta(z; \lambda)) \right\}. \quad (90)$$

Because $d_m(r) = \mathcal{O}(r^m)$, the expression in braces is a formal power series in z , so Equation (90) is well-defined. It gives

$$\frac{d_m(r)}{n^m} \sim \frac{L^m}{n^m} \sum_{j \geq 0} \frac{p_{m,j}(\lambda)}{L^j}. \quad (91)$$

Every $p_{m,j}$ is obtained by a finite computation from Stirling numbers of both kinds, Gaussian moments, and ordinary coefficient extraction.

For illustration, the first sector begins

$$\begin{aligned} \frac{d_1(r)}{n} \sim \frac{L}{n} \left(-\frac{1}{12} + \frac{2\lambda - 1}{24L} - \frac{2\lambda + 1}{24L^2} - \frac{\lambda^2 - \lambda - 8}{24L^3} \right. \\ \left. - \frac{2\lambda^3 - 6\lambda^2 - 45\lambda + 60}{72L^4} + \dots \right). \end{aligned} \quad (92)$$

The next two begin

$$\frac{d_2(r)}{n^2} \sim \frac{L^2}{n^2} \left(-\frac{1}{24L} + \frac{\lambda}{24L^2} - \frac{2\lambda - 1}{48L^3} - \frac{\lambda^2 - 3\lambda + 6}{48L^4} + \dots \right), \quad (93)$$

$$\frac{d_3(r)}{n^3} \sim \frac{L^3}{n^3} \left(\frac{1}{360} - \frac{12\lambda - 5}{1440L} + \frac{12\lambda^2 + 2\lambda - 35}{1440L^2} + \dots \right). \quad (94)$$

7.5 The full polynomial-logarithmic transseries

Combining Equations (82), (88) and (91) gives the promised all-level expansion.

Theorem 7.1 (Full logarithmic transseries in L and λ). *With $L = \log n$, $\lambda = \log L$,*

$$\log \mathcal{B}_n \sim n \left(L - \lambda - 1 + \sum_{N \geq 1} \frac{H_N(\lambda)}{L^N} \right) - 1 - \frac{1}{2} \log L + \sum_{N \geq 1} \frac{E_N(\lambda)}{L^N} + \sum_{m \geq 1} \frac{L^m}{n^m} \sum_{j \geq 0} \frac{p_{m,j}(\lambda)}{L^j}. \quad (95)$$

The coefficient families are given explicitly by [Equations \(78\), \(81\), \(87\) and \(90\)](#), while the d_m entering [Equation \(90\)](#) are defined by the finite formulae [Equations \(29\), \(49\) and \(51\)](#).

Proof. The first two lines are formal composition of [Equation \(77\)](#) with the rational functions in [Equation \(52\)](#). Because the inner series has leading monomial L , inversion, logarithm, and rational composition are valid in the polynomial-logarithmic transseries ring. The coefficient extraction [Equation \(90\)](#) handles each exponentially small sector separately. The hierarchy is well ordered by first comparing the power of n^{-1} , then the power of L^{-1} within that sector. $\square$

Remark 7.2 (Exponentiating the full logarithmic transseries). [Equation \(95\)](#) is usually the cleanest form because $\mathcal{B}_n$ is enormous. The Bell number itself is obtained simply as $\mathcal{B}_n \sim \exp(\text{right-hand side})$. Alternatively, exponentiate the last line sector by sector using complete Bell polynomials; this reproduces the bounded multiplicative coefficients A_m of [Equation \(70\)](#) after the substitution $r = W_0(n)$.

8 Nonprincipal saddles and hyperasymptotic sectors

8.1 Universal local germs

The phase $\Psi_n(z) = e^z - 1 - n \log z$ has saddles $r_k = W_k(n)$. Around any nondegenerate saddle, the same local calculation applies after analytic continuation. Thus the formal local germ is

$$\mathfrak{S}_k(n) \sim \frac{n! \exp(e^{r_k} - 1)}{r_k^n \sqrt{2\pi n(1 + r_k)}} \sum_{m \geq 0} \frac{c_m(r_k)}{n^m}, \quad (96)$$

with a consistent branch of the square root chosen along the steepest-descent direction. The coefficient functions c_m need no modification: they are rational and hence analytically continue away from $r_k = -1$.

8.2 Relative actions

Let

$$\Delta_k(n) := \Psi_n(r_k) - \Psi_n(r_0). \quad (97)$$

For fixed k , put $y = 2\pi i k$. The relation $r_k + \log r_k = r + \log r + y$ may be written

$$\delta_k + \log \left(1 + \frac{\delta_k}{r} \right) = y, \quad \delta_k := r_k - r. \quad (98)$$

Formal reversion gives

$$\delta_k = y - \frac{y}{r} + \frac{y(y+2)}{2r^2} + \frac{y(-2y^2 - 9y - 6)}{6r^3} + \mathcal{O}_k(r^{-4}). \quad (99)$$

Using $e^r = n/r$, one obtains

$$\frac{\Delta_k(n)}{n} = -\frac{y}{r} + \frac{y^2}{2r^2} - \frac{y^2/2 + y^3/3}{r^3} + \mathcal{O}_k(r^{-4}). \quad (100)$$

Hence

$$\operatorname{Re} \Delta_k(n) = -\frac{2\pi^2 k^2 n}{r^2} \left(1 - \frac{1}{r} + \mathcal{O}_k(r^{-2})\right). \quad (101)$$

The adjacent complex saddles are therefore suppressed on the scale

$$\exp\left(-\frac{2\pi^2 k^2 n}{r^2}(1 + \mathcal{O}_k(r^{-1}))\right), \quad (102)$$

which is smaller than e^{-Mr} for every fixed M .

Why no unconditional global saddle sum is asserted

A formal expression

$$\sum_{k \in \mathbb{Z}} \sigma_k(n) \mathfrak{S}_k(n)$$

requires the intersection numbers or Stokes multipliers $\sigma_k(n)$ of the original Cauchy contour with the steepest-descent thimbles. The local coefficient calculus determines $\mathfrak{S}_k$, but not those global multipliers. For the ordinary positive-real Poincaré expansion, $k = 0$ alone suffices and the remainder theorem of [Theorem 3.3](#) is unconditional. The scales [Equation \(102\)](#) describe the natural hyperasymptotic remainder and the location of possible exponentially tiny oscillations, not an unproved exact transseries decomposition.

9 Touchard polynomials and shifted Bell numbers

9.1 Universality for $\mathcal{B}_n(x)$

Define the Touchard polynomial

$$\exp(x(e^z - 1)) = \sum_{n \geq 0} \mathcal{B}_n(x) \frac{z^n}{n!}, \quad \mathcal{B}_n(x) = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^k. \quad (103)$$

For fixed $x > 0$, the positive saddle satisfies

$$xre^r = n, \quad r = W_0(n/x). \quad (104)$$

The phase derivatives are $xe^r \mathcal{T}_j(r)$. Since $xe^r = n/r$, the normalized quantities $q_j(r)$ are exactly the same as before.

Theorem 9.1 (Universal Touchard expansion). *Let $K \subset (0, \infty)$ be compact. Uniformly for $x \in K$, with $r = W_0(n/x)$,*

$$\mathcal{B}_n(x) = \frac{n! \exp(x(e^r - 1))}{r^n \sqrt{2\pi n(1+r)}} \left(\sum_{m=0}^{M-1} \frac{c_m(r)}{n^m} + \mathcal{O}_{M,K}((r/n)^M) \right). \quad (105)$$

The same c_m from [Equation \(29\)](#) works for every x . After Stirling's expansion,

$$\log \mathcal{B}_n(x) = x(e^r - 1) + nr - n - \frac{1}{2} \log(1+r) + \sum_{m=1}^{M-1} \frac{d_m(r)}{n^m} + \mathcal{O}_{M,K}((r/n)^M). \quad (106)$$

Proof. Repeat Equation (6) with $e^{re^{i\theta}} - e^r$ multiplied by x . Equation (104) gives the same quadratic coefficient $A = n(1+r)$ and the same scaled phase Equation (16). The localization estimates are uniform for x in a compact positive interval because $r = W_0(n/x) \sim \log n$ uniformly there. The logarithmic form follows as in Theorem 6.2. $\square$

Remark 9.2. The universality is stronger than a mere substitution. The parameter x changes the saddle location and the leading exponential, but after the saddle equation is imposed it disappears from every standardized local cumulant. Regimes in which $x = x_n \rightarrow 0$ or $x_n \rightarrow \infty$ require separate uniformity conditions; Theorem 9.1 intentionally treats fixed positive parameter ranges.

9.2 A fixed-shift coefficient formula

For fixed integer h , one may either apply the main theorem directly to $N = n + h$, or keep the saddle $r = W_0(n)$. The latter gives

$$\frac{\mathcal{B}_{n+h}}{(n+h)!} = \frac{\exp(e^r - 1)}{2\pi r^{n+h}} \int_{-\pi}^{\pi} \exp(\Phi_r(\theta)) e^{-ih\theta} d\theta. \quad (107)$$

After the Gaussian scaling, define

$$c_m(r; h) := \mathbb{E} \left[t^{2m} \exp \left(-\frac{ihZ}{\sqrt{1+r}} t + \sum_{s \geq 1} a_s(Z; r) t^s \right) \right]. \quad (108)$$

Then, uniformly for $|h| \leq H$ with fixed H ,

$$\mathcal{B}_{n+h} = \frac{(n+h)! \exp(e^r - 1)}{r^{n+h} \sqrt{2\pi n(1+r)}} \left(\sum_{m=0}^{M-1} \frac{c_m(r; h)}{n^m} + \mathcal{O}_{M,H}((r/n)^M) \right). \quad (109)$$

Thus the fixed-shift coefficients are again finite complete Bell polynomials; the only new atom is the linear Gaussian term in Equation (108).

10 Algorithms and exact coefficient generation

The formulae above are not merely existential. They give several independent coefficient engines, each suited to a different task. This section records the engines in a form that can be implemented in an exact computer-algebra system or formalized over the rational-function field $\mathbb{Q}(r)$.

10.1 A radical-free partition algorithm

Although the standardized atoms $q_j(r)$ in Equation (15) contain half-integer powers of $1+r$, every contributing monomial in Equation (29) has an even total Gaussian degree. Define

$$U_j(r) := \frac{\mathcal{T}_j(r)}{r} = \sum_{k=1}^j \left\{ \begin{matrix} j \\ k \end{matrix} \right\} r^{k-1} \in \mathbb{N}[r]. \quad (110)$$

For a vector $\mathbf{k} = (k_1, \dots, k_{2m})$ satisfying $\sum_{s=1}^{2m} s k_s = 2m$, put $K = \sum_s k_s$. Since

$$\frac{1}{2} \sum_{s=1}^{2m} (s+2) k_s = m + K,$$

Equation (29) becomes the completely radical-free identity

$$c_m(r) = \sum_{\substack{k_1, \dots, k_{2m} \geq 0 \\ \sum_{s=1}^{2m} s k_s = 2m}} \frac{(-1)^{m+K} (2m + 2K - 1)!!}{(1+r)^{m+K}} \prod_{s=1}^{2m} \frac{U_{s+2}(r)^{k_s}}{((s+2)!)^{k_s} k_s!}. \quad (111)$$

This is often the cleanest exact formula. It requires one term for each integer partition of $2m$, not one term for each composition. In particular, it is finite before any simplification, and all of its arithmetic takes place in $\mathbb{Q}(r)$.

Algorithm 10.1 (Partition engine for c_m). Given $m \geq 0$:

1. Generate $\mathcal{T}_0 = 1$ and $\mathcal{T}_{j+1} = r(\mathcal{T}_j + \mathcal{T}'_j)$ up to $j = 2m + 2$.
2. Set $U_j = \mathcal{T}_j/r$ for $2 \leq j \leq 2m + 2$.
3. Enumerate the multiplicity vectors $\mathbf{k}$ with $\sum s k_s = 2m$.
4. Add the summand in Equation (111) for each vector.
5. Put the result over the common denominator $(1+r)^{3m}$, cancel common factors, and return the numerator polynomial.

For $m = 0$, return $c_0 = 1$ without entering the loop.

The exponent $m + K$ never exceeds $3m$, because $K \leq 2m$. This makes the denominator theorem Theorem 4.4 visible directly in the algorithm. The degree estimate follows just as transparently: the product of the U_{s+2} 's has degree at most $\sum (s+1)k_s = 2m + K$, and multiplication by $(1+r)^{3m-(m+K)}$ raises it to at most $4m$.

10.2 Dynamic Bell-polynomial algorithm

Partition enumeration is conceptually direct, while the recurrence Equation (30) is usually faster for moderate orders. Let $\mathcal{G} : \mathbb{Q}(r)[u] \rightarrow \mathbb{Q}(r)$ be the Gaussian moment functional

$$\mathcal{G}(u^{2j}) = (2j-1)!!, \quad \mathcal{G}(u^{2j+1}) = 0. \quad (112)$$

Construct $a_s(u; r)$ from Equation (16), then compute

$$P_0(u; r) = 1, \quad P_j(u; r) = \frac{1}{j} \sum_{s=1}^j s a_s(u; r) P_{j-s}(u; r). \quad (113)$$

The desired answer is $c_m = \mathcal{G}(P_{2m})$. Only $a_1, \dots, a_{2m}$ are needed. Odd values of P_j need not be Gaussian-averaged, but they must be retained because they feed later even values.

Exact symbolic pseudocode

The following pseudocode uses the recurrence, but applies the Gaussian functional before the final rational simplification. The symbol `stirling2(j, k)` denotes a Stirling number of the second kind.

```
def touchard(j, r):
    return sum(stirling2(j, k) * r**k for k in range(1, j + 1))
```

```

def gaussian_moment_polynomial(poly, u):
    ans = 0
    for exponent, coefficient in coefficients_in(poly, u):
        if exponent % 2 == 0:
            ans += coefficient * double_factorial(exponent - 1)
    return ans

def saddle_coefficient(m, r, u):
    if m == 0:
        return 1
    a = {}
    for s in range(1, 2*m + 1):
        T = touchard(s + 2, r)
        q = T / (r * (1 + r)**((s + 2)/2))
        a[s] = I**(s + 2) * q * u**(s + 2) / factorial(s + 2)
    P = {0: 1}
    for j in range(1, 2*m + 1):
        P[j] = expand(sum(s*a[s]*P[j-s] for s in range(1, j + 1))/j)
    return factor(gaussian_moment_polynomial(P[2*m], u))

```

In exact software, the exponent $(s + 2)/2$ should be represented as a rational number, not as floating-point division. Alternatively, use [Equation \(111\)](#) and avoid radicals altogether.

10.3 Logarithmic, exponential, and polynomial-logarithmic engines

Once $c_1, \dots, c_M$ are known, no further integration is needed. The remaining coefficient systems are ordinary formal-series transforms. For convenient reference, the pipeline is

$$\{c_m\} \xrightarrow{\log} \{\ell_m\} \xrightarrow{+\sigma_m} \{d_m\} \xrightarrow{d_m/r^m} \{D_m\} \xrightarrow{\exp} \{A_m\}. \quad (114)$$

The arrows are implemented respectively by [Equations \(50\), \(51\), \(55\) and \(69\)](#). Explicitly,

$$\ell_m = c_m - \frac{1}{m} \sum_{j=1}^{m-1} j \ell_j c_{m-j}, \quad (115)$$

$$d_m = \ell_m + \sigma_m, \quad D_m = \frac{d_m}{r^m}, \quad (116)$$

$$A_0 = 1, \quad A_m = \frac{1}{m} \sum_{j=1}^m j D_j A_{m-j}. \quad (117)$$

All operations are triangular: coefficients through order M require only coefficients through order M at the preceding stage.

For the polynomial-logarithmic sector, first compute the Lambert polynomials $R_N(\lambda)$ by [Equation \(78\)](#), then build the truncated formal series

$$\Delta_M(z) = 1 - \lambda z + \sum_{j=1}^{M-1} R_j(\lambda) z^{j+1}. \quad (118)$$

Ordinary truncated multiplication, reciprocal, and logarithm operations give H_N from [Equation \(81\)](#), E_N from [Equation \(87\)](#), and every $p_{m,j}$ from [Equation \(90\)](#). Thus the full transseries is generated by three familiar coefficient calculi: Stirling-cycle numbers for W , complete Bell polynomials for composition and exponentiation, and Gaussian moments for the saddle integral.

10.4 Complexity and internal checks

Let $p(N)$ be the integer partition function. The direct formula for c_m has $p(2m)$ summands. The dynamic recurrence has only $O(m^2)$ formal convolutions, although polynomial expansion and rational simplification dominate the actual bit complexity. In either approach, exact coefficient generation is practical well beyond the four orders printed in this article.

A reliable implementation should enforce the following invariants at every order:

- (i) $(1 + r)^{3m} c_m(r) \in \mathbb{Q}[r]$;
- (ii) its degree is at most $4m$;
- (iii) $c_m(r) = \mathcal{O}(r^m)$ as $r \rightarrow \infty$;
- (iv) $D_m(r)$ has a finite limit given by [Equation \(74\)](#);
- (v) the limiting A_m 's satisfy the inverse Stirling identity [Equation \(37\)](#);
- (vi) numerical substitution into [Equation \(23\)](#) improves the relative error by approximately one factor of $r/n = e^{-r}$ per retained order.

These checks detect sign errors, missing factorials, and incorrect Gaussian moments very efficiently.

11 Numerical validation

11.1 Definition of the tested approximants

For $M \geq 0$, define the principal-saddle approximant

$$S_M(n) := \frac{n! \exp(e^r - 1)}{r^n \sqrt{2\pi n(1+r)}} \sum_{m=0}^M \frac{c_m(r)}{n^m}, \quad r = W_0(n). \quad (119)$$

Thus S_0 is the classical leading approximation and S_4 uses c_1, c_2, c_3, c_4 , with c_4 printed in [Section A](#). Exact Bell numbers were generated by

$$\mathcal{B}_0 = 1, \quad \mathcal{B}_{n+1} = \sum_{k=0}^n \binom{n}{k} \mathcal{B}_k, \quad (120)$$

and all transcendental evaluations were performed at substantially higher precision than the digits shown.

11.2 Relative-error table

The table reports $\varepsilon_M(n) := S_M(n)/\mathcal{B}_n - 1$. The signs are retained because they provide a more sensitive check than absolute errors alone.

The data exhibit the predicted Poincaré ordering. Because $r/n = e^{-r}$, the nominal ratio between consecutive terms is of order r/n , multiplied by a coefficient that grows polynomially in r . The improvement is already substantial at $n = 10$, and by $n = 500$ four corrections reduce the relative error to roughly $2.5 \cdot 10^{-14}$. The small nonmonotonicity between orders two and three at $n = 10$ is normal for a finite asymptotic expansion and does not signal a failure of the coefficient formula.

Table 1: Relative errors of the all-orders principal-saddle approximants.

n	$M = 0$	$M = 1$	$M = 2$	$M = 3$	$M = 4$
10	$2.6771612 \cdot 10^{-2}$	$3.8253603 \cdot 10^{-4}$	$-1.2965341 \cdot 10^{-5}$	$-6.2268398 \cdot 10^{-6}$	$-1.1599281 \cdot 10^{-6}$
20	$1.5326828 \cdot 10^{-2}$	$1.1293364 \cdot 10^{-4}$	$-3.8237417 \cdot 10^{-6}$	$-7.5548761 \cdot 10^{-7}$	$-7.1302490 \cdot 10^{-8}$
50	$7.2381178 \cdot 10^{-3}$	$2.1197165 \cdot 10^{-5}$	$-5.8160607 \cdot 10^{-7}$	$-4.1043863 \cdot 10^{-8}$	$-1.3992028 \cdot 10^{-9}$
100	$4.0615707 \cdot 10^{-3}$	$5.7437480 \cdot 10^{-6}$	$-1.2399466 \cdot 10^{-7}$	$-4.2053807 \cdot 10^{-9}$	$-6.2657058 \cdot 10^{-11}$
200	$2.2599385 \cdot 10^{-3}$	$1.5057677 \cdot 10^{-6}$	$-2.4589871 \cdot 10^{-8}$	$-4.0769478 \cdot 10^{-10}$	$-2.4569360 \cdot 10^{-12}$
500	$1.0289535 \cdot 10^{-3}$	$2.4364162 \cdot 10^{-7}$	$-2.6649303 \cdot 10^{-9}$	$-1.7364703 \cdot 10^{-11}$	$-2.4871987 \cdot 10^{-14}$

11.3 Independent check of the first correction

Using $n = re^r$, the logarithmic first-order coefficient gives

$$\exp\left(\frac{d_1(r)}{n} + \mathcal{O}((r/n)^2)\right) = 1 - \frac{r^2(2r^2 + 7r + 10)}{24(1+r)^3n} + \mathcal{O}\left(\frac{r^2}{n^2}\right). \quad (121)$$

Consequently,

$$\mathcal{B}_n = \frac{\exp(n(r - 1 + r^{-1}) - 1)}{\sqrt{1+r}} \left(1 - \frac{r^2(2r^2 + 7r + 10)}{24(1+r)^3n} + \mathcal{O}\left(\frac{r^2}{n^2}\right)\right), \quad (122)$$

which is algebraically identical to the independently obtained 2026 first-correction formula discussed in [Section 6](#). This comparison is important: it tests simultaneously the saddle coefficient c_1 , the Stirling correction $1/(12n)$, and the conversion from the factorial form to the Lambert–W exponential form.

12 Proof audit, transseries meaning, and limitations

12.1 What has been proved

The principal result is an ordinary asymptotic theorem with a moving parameter. For every fixed M , the coefficients are evaluated at $r = W_0(n)$, and [Theorem 3.3](#) states a uniform remainder of relative size $\mathcal{O}_M((r/n)^M)$ after M retained coefficients. Since $r \sim \log n$, [Theorem 4.5](#) proves that these terms form a genuine asymptotic sequence. The finite coefficient formulae are exact identities in $\mathbb{Q}(r)$, not asymptotic guesses.

The following implications are therefore rigorous:

$$\text{localized Cauchy integral} \implies \{c_m\} \implies \{d_m, D_m, A_m\} \implies \text{every finite truncation in } r. \quad (123)$$

The polynomial–logarithmic formula then follows by composing any prescribed finite truncation with the convergent large-positive-argument expansion of W_0 , or formally to all orders by coefficient extraction. At a fixed requested precision, only finitely many W -coefficients and finitely many saddle coefficients contribute.

12.2 The ranking of the full transseries

The phrase “full transseries” can be ambiguous unless the ordering is stated. The expansion constructed here has three nested ranks:

1. The *extensive rank*, of size n , is the expansion of $n(r - 1 + r^{-1})$ in powers of L^{-1} with polynomial coefficients in $\lambda = \log L$.

2. The *edge rank*, of size one, comes from $-1 - \frac{1}{2} \log(1+r)$ and has its own $L^{-1}\mathbb{Q}[\lambda]$ -series.
3. The *exponentially small ranks* are indexed by $m \geq 1$ and have magnitude $e^{-mr} = n^{-m} L^m (1 + o(1))$, each carrying a complete $L^{-1}\mathbb{Q}[\lambda]$ -expansion.

Within a fixed exponential sector, powers of L^{-1} give the local ordering. Across sectors, every term of the $m+1$ sector is beyond all algebraic orders of the m sector after a finite polynomial-logarithmic truncation. This is the precise sense in which Equation (95) is a transseries rather than one undifferentiated formal series.

12.3 What is not claimed

Boundaries of the present result

- (a) No convergence claim is made for the infinite principal-saddle series $\sum c_m(r) n^{-m}$ when the order tends to infinity together with n . The theorem is Poincaré asymptotic: M is fixed first.
- (b) No sharp factorial-over-power estimate for c_m as $m \rightarrow \infty$ is proved. Such an estimate is needed for rigorous optimal truncation and resurgence.
- (c) The formal nonprincipal saddle germs in Section 8 do not by themselves determine which complex saddles occur on a deformation of the original Cauchy circle, nor their Stokes multipliers.
- (d) The Touchard theorem is uniform only for x in a fixed compact subset of $(0, \infty)$. Coalescing regimes such as $x \rightarrow 0$, or complex x near Stokes curves, require a separate analysis.
- (e) The document supplies a formalization map but not a machine-checked proof. In particular, the complex-analytic localization estimates and all truncation bookkeeping would have to be encoded explicitly in a proof assistant.

These qualifications are mathematically substantive. They separate the complete all-orders coefficient calculus proved here from a stronger, still open hyperasymptotic description of the exact Bell sequence.

12.4 A formalization map

The proof naturally decomposes into modules.

Module	Principal statements and dependencies
Combinatorial algebra	Stirling recurrence; Touchard recurrence Equation (11); divisibility $r \mid \mathcal{T}_j$; complete Bell polynomials; integer-partition indexing.
Formal power series	Exponential recurrence Equation (30); logarithm recurrence Equation (50); exponential recurrence Equation (69); truncated composition and inversion.
Gaussian calculus	Moment identity Equation (112); Wick pairing; equivalence of Equations (27) and (29).
Real asymptotics	Bounds for $r = W_0(n)$; estimates Equations (18) and (19); domination of Taylor remainders by a Gaussian integrable majorant.
Complex analysis	Cauchy's coefficient formula; parameterization of the saddle circle; complementary-arc localization.

Module	Principal statements and dependencies
Lambert calculus	Stirling-cycle-number expansion Equation (78) ; finite coefficient extraction for $H_N, E_N, p_{m,j}$.

The algebraic modules can be developed independently of analysis and reused for other admissible exponential generating functions. The only Bell-specific analytic input is the phase $e^{re^{i\theta}}$, whose derivatives are encoded by the Touchard polynomials.

13 Further deductions and research directions

The coefficient calculus reveals several questions that are not visible at leading order.

13.1 Large-order growth and resurgence

The nearest competing saddles are $W_{\pm 1}(n)$. Their action gap is of magnitude $2\pi^2 n/r^2$ by [Equation \(101\)](#). Standard resurgence heuristics therefore suggest that the late terms of the principal-saddle series should eventually be controlled by these adjacent actions. A natural program is to prove a bound of schematic form

$$c_m(r) \sim \sum_{k=\pm 1} \frac{S_k(r) \Gamma(m + \alpha_k)}{\Delta_k(r)^{m+\alpha_k}} \left(1 + \frac{b_{k,1}(r)}{m} + \dots \right), \quad (124)$$

with explicitly determined Stokes constants S_k . Formula [Equation \(29\)](#) supplies arbitrarily many exact low-order data with which such a conjecture can be tested, but it does not prove [Equation \(124\)](#).

13.2 Ratio, logarithmic derivative, and finite differences

Because all transforms are formal and triangular, the present results immediately generate complete expansions of

$$\frac{\mathcal{B}_{n+1}}{\mathcal{B}_n}, \quad \log \mathcal{B}_{n+1} - \log \mathcal{B}_n, \quad \Delta^j \log \mathcal{B}_n, \quad \frac{\mathcal{B}_{n+h}}{\mathcal{B}_n} \quad (125)$$

for fixed j, h . There are two equivalent routes: substitute $n \mapsto n + h$ into the master transseries and re-expand, or use the shifted coefficient germ [Equation \(108\)](#). Comparing the two gives an infinite family of exact identities among the rational functions $c_m(r; h)$, the derivatives of W , and the unshifted $c_m(r)$. These identities appear well suited to automated proof and can sharpen explicit ratio bounds.

13.3 Uniform Touchard regimes

The universality theorem suggests several boundary problems:

- (i) $x = x_n \rightarrow 0^+$, where $r = W(n/x)$ grows faster than $\log n$;
- (ii) $x = x_n \rightarrow \infty$, where the saddle can leave the logarithmic regime;
- (iii) negative or complex x , where several saddles can have comparable real action;
- (iv) double scaling near a Stokes curve, requiring an error-function or Airy-type smoothing rather than a single Gaussian saddle.

The finite local coefficients remain meaningful by analytic continuation, but the global contour and uniform remainder must be rebuilt in each regime.

13.4 A general exponential-of-exponential template

Consider a family

$$F(z) = \exp(H(z)), \quad [z^n]F(z) = \frac{F(r)}{r^n \sqrt{2\pi b(r)}} (1 + \dots), \quad (126)$$

where $rH'(r) = n$ and $b(r) = (r\partial_r)^2 H(r)$. Define normalized logarithmic derivatives

$$\kappa_j(r) := \frac{(r\partial_r)^j H(r)}{b(r)^{j/2}}. \quad (127)$$

Exactly the same Gaussian Bell-polynomial calculation gives every local saddle coefficient as a finite polynomial in $\kappa_3, \dots, \kappa_{2m+2}$. Bell numbers are special because $H(z) = e^z - 1$ converts these derivatives into Touchard polynomials and the saddle into a Lambert W -function. This observation provides a systematic route from the repository's combinatorial coefficient calculus to a wider class of labelled combinatorial assemblies.

14 Conclusion

The Bell-number saddle is classical; the point of this article is to make its entire correction mechanism explicit and finite. The result can be summarized in four layers.

First, all local derivatives of the phase are Touchard polynomials. Second, Gaussian reduction turns those derivatives into the exact finite coefficient formula [Equation \(29\)](#), or equivalently the radical-free form [Equation \(111\)](#). Third, ordinary logarithm and exponential recurrences combine the saddle series with Stirling's series to produce the bounded rational coefficient families $D_m(r)$ and $A_m(r)$ in the natural exponential variable $e^{-r} = r/n$. Fourth, the Stirling-cycle-number expansion of $W_0(n)$ resolves every one of these coefficients into the ranked polynomial-logarithmic transseries in $L = \log n$ and $\lambda = \log L$.

Thus every coefficient at every displayed rank is governed by a finite formula. The theorem includes an arbitrary-order principal-saddle remainder, its first correction agrees with an independent recent derivation, and direct numerics confirm rapid improvement. The remaining frontier is no longer the local coefficient calculus; it is the global hyperasymptotic problem of nonprincipal saddles, large-order coefficient growth, and Stokes switching.

A The fourth principal-saddle coefficient

For numerical checks and as a higher-order checksum, the fourth coefficient is

$$c_4(r) = -\frac{N_4(r)}{39813120(1+r)^{12}}, \quad (128)$$

where

$$\begin{aligned} N_4(r) = & 9136r^{16} - 357408r^{15} - 5092600r^{14} - 29432904r^{13} - 98499653r^{12} \\ & - 212369184r^{11} - 302729704r^{10} - 277448328r^9 - 156972248r^8 \\ & - 52300272r^7 + 110053352r^6 + 302787360r^5 + 198352r^4 \\ & + 2545056r^3 + 647456r^2 + 109632r + 9136. \end{aligned}$$

The sign in Equation (128) applies to the whole numerator. In particular, the leading behavior is

$$c_4(r) = -\frac{571}{2488320}r^4 + \mathcal{O}(r^3), \quad (129)$$

in agreement with $\gamma_4 = -571/2488320$ in Equation (39).

B Coefficient dictionaries

This appendix collects the coefficient-extraction rules in one place. They are useful when translating between notational conventions in the literature.

B.1 Principal saddle

With $Z \sim N(0, 1)$,

$$a_s(Z; r) = \frac{i^{s+2} \mathcal{J}_{s+2}(r)}{(s+2)! r(1+r)^{(s+2)/2}} Z^{s+2}, \quad (130)$$

$$c_m(r) = \mathbb{E}[t^{2m}] \exp\left(\sum_{s \geq 1} a_s(Z; r) t^s\right). \quad (131)$$

Equations (130) and (131) are the shortest universal definition; Equation (111) is the shortest radical-free definition.

B.2 Logarithmic and multiplicative sectors

Let $C(x) = 1 + \sum_{m \geq 1} c_m x^m$. Then

$$\log C(x) = \sum_{m \geq 1} \ell_m x^m, \quad d_m = \ell_m + \sigma_m, \quad D_m = r^{-m} d_m, \quad (132)$$

$$\exp\left(\sum_{m \geq 1} D_m y^m\right) = \sum_{m \geq 0} A_m y^m. \quad (133)$$

The Bell number can therefore be written either as

$$\log \mathcal{B}_n \sim n(r - 1 + r^{-1}) - 1 - \frac{1}{2} \log(1 + r) + \sum_{m \geq 1} D_m(r) e^{-mr}, \quad (134)$$

$$\mathcal{B}_n \sim \frac{\exp(n(r - 1 + r^{-1}) - 1)}{\sqrt{1 + r}} \sum_{m \geq 0} A_m(r) e^{-mr}. \quad (135)$$

B.3 Inverse- r and polynomial-logarithmic subcoefficients

Because every $D_m(r)$ is regular at $r = \infty$, define

$$D_{m,j} := [u^j] D_m(u^{-1}), \quad D_m(r) = \sum_{j \geq 0} D_{m,j} r^{-j} \quad (136)$$

formally at infinity. Likewise $A_{m,j} := [u^j] A_m(u^{-1})$. Substitution of $r = L\Delta(L^{-1})$ gives the sector coefficients

$$p_{m,j}(\lambda) = [z^j] [z^m d_m(z^{-1} \Delta(z))] = [z^j] [\Delta(z)^m D_m(z^{-1} \Delta(z))]. \quad (137)$$

This second equality makes explicit that the leading value is controlled by $D_m(\infty)$, whereas all lower inverse powers of r generate the internal L^{-1} -tail of the same exponential sector.

C A direct derivation of the first coefficient

For readers wishing to audit every normalization, retain terms through order n^{-1} in [Equation \(16\)](#):

$$\exp\left(n^{-1/2}a_1 + n^{-1}a_2 + \mathcal{O}(n^{-3/2})\right) = 1 + n^{-1/2}a_1 + n^{-1}\left(a_2 + \frac{1}{2}a_1^2\right) + \mathcal{O}(n^{-3/2}). \quad (138)$$

The odd term has zero Gaussian mean. Since

$$a_1 = -\frac{iq_3}{6}Z^3, \quad a_2 = \frac{q_4}{24}Z^4,$$

we obtain

$$c_1 = \frac{q_4}{24}\mathbb{E}Z^4 - \frac{q_3^2}{72}\mathbb{E}Z^6 = \frac{q_4}{8} - \frac{5q_3^2}{24}. \quad (139)$$

Now

$$\mathcal{T}_3(r) = r + 3r^2 + r^3, \quad \mathcal{T}_4(r) = r + 7r^2 + 6r^3 + r^4,$$

so substitution and collection over $24(1+r)^3$ yield

$$c_1(r) = -\frac{2r^4 + 9r^3 + 16r^2 + 6r + 2}{24(1+r)^3}.$$

Finally, adding the first Stirling logarithm coefficient gives

$$\begin{aligned} d_1(r) &= c_1(r) + \frac{1}{12} \\ &= -\frac{r^2(2r^2 + 7r + 10)}{24(1+r)^3}, \end{aligned}$$

which explains the cancellation of the constant and linear terms in the numerator. This cancellation is the first visible instance of the Bernoulli boundary mechanism in [Theorem 6.3](#).

D Bibliographic notes

Moser and Wyman identified the Lambert-type saddle and developed classical correction expansions. Hayman's admissibility theory places the Bell generating function in a broad coefficient-asymptotic framework, while de Bruijn gives the familiar logarithmic form. Harris and Schoenfeld developed general coefficient expansions for analytic functions. Corless et al. and the DLMF provide the branch-aware all-orders Lambert W expansion used in [Section 7](#). More recent work supplies explicit bounds and an independent elementary derivation of the first Bell correction.

The novelty claimed here is organizational and algebraic rather than a claim that saddle-point expansions for Bell numbers were previously unknown: the article makes every coefficient finite and computable, proves the rational structure adapted to $r = W(n)$, and threads the result through logarithmic, exponential, and polynomial-logarithmic transseries in one consistent notation.

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